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ETHNOTECH ACADEMIC SOLUTIONS
DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEM

Report on

SMART LIBRARY INVENTORY & ACCESS MANAGEMENT

Report submitted for the award of Assignment marks of Subject Name: Full Stack Web Development

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• ABSTRACT

This project presents the design and development of a Smart Library Inventory & Access Management System, a full-stack web application aimed at digitalizing and streamlining the operations of a library. The primary objective is to create an efficient, user-friendly, and secure platform that manages book inventories, member registrations, book issue/return workflows, and access control.

The system supports multiple user roles including Administrator, Librarian, and Members. Administrators can manage the entire library inventory, track issued books, generate reports, and control user access. Librarians can add, update, or remove books, manage member records, and process book transactions. Members can browse the catalog, view availability, and track their borrowing history.

The project is implemented using Python (Flask framework) for the backend, HTML, CSS, and JavaScript for the frontend, and MySQL for persistent data storage. The outcome delivers a scalable and maintainable library management solution that eliminates manual record-keeping and improves operational efficiency.

• INTRODUCTION

Libraries are essential knowledge centers that house thousands of books and resources. However, managing these resources manually has long been a challenge — prone to errors, delays, and inefficiencies. With the rise of digital technologies, library management systems have become an indispensable tool for educational institutions and public libraries alike.

The Smart Library Inventory & Access Management System is developed to address these challenges by automating key library processes. This system enables real-time tracking of book availability, member activity, and due dates. It ensures that administrators and librarians have complete visibility and control over library operations, while members enjoy a seamless browsing and borrowing experience.

This project focuses on building a robust full-stack application using Python, HTML, CSS, JavaScript, and MySQL — technologies widely adopted in modern web development for their reliability, flexibility, and scalability.

• PROBLEM STATEMENT

Traditional library management systems rely heavily on manual record-keeping, which leads to issues such as misplaced books, incorrect issue/return records, delayed fine calculations, and lack of real-time inventory visibility.

There is a need for a centralized, automated, and role-based digital system that manages book inventory, member access, and library transactions efficiently. The Smart Library Inventory & Access Management System aims to solve these challenges by providing a comprehensive, database-driven web application.

• OBJECTIVES OF THE PROJECT

The key objectives of this project are:

- To build a centralized digital platform for library inventory management
- To implement role-based access control for Admin, Librarian, and Members
- To automate book issue, return, and fine calculation processes
- To maintain real-time tracking of book availability and member borrowing history
- To provide a searchable and filterable book catalog for members
- To generate inventory and transaction reports for administrators
- To store all library data securely using a relational MySQL database

• SCOPE OF THE PROJECT

The scope of the project includes the following modules and functionalities:

- Admin dashboard for full system control and reporting
- Book inventory management (Add, Update, Delete, Search books)
- Member registration and profile management
- Book issue and return workflow with due date tracking
- Fine calculation for overdue books
- Role-based login and access control
- MySQL database for all persistent data storage

The project does not include integration with external library networks, mobile applications, or RFID-based tracking. However, the architecture is designed to support such extensions in the future.

• LITERATURE SURVEY

Existing library management systems such as Koha, OpenBiblio, and OPAC-based platforms provide comprehensive digital solutions for managing library resources. These systems are known for their:

- Centralized inventory and cataloging features
- Member management and borrowing history tracking
- Fine calculation and notification systems
- Report generation and analytics dashboards

The study of these platforms highlights the importance of a well-structured database, efficient search functionality, and secure role-based access. The Smart Library project draws inspiration from these systems while simplifying the implementation using open-source technologies suitable for academic deployment.

• SYSTEM REQUIREMENTS

Hardware Requirements

- Laptop or Desktop Computer
- Minimum 8 GB RAM
- Internet connection (for deployment and browser access)

Software Requirements

- Operating System: Windows / Linux / macOS
- Python 3.x
- Flask Framework
- MySQL Server
- Visual Studio Code
- Web Browser (Google Chrome / Firefox)

TECHNOLOGY STACK

- Python (Flask): Backend server, routing, and business logic
- HTML5 / CSS3: Frontend structure and styling
- JavaScript: Dynamic UI behavior and AJAX calls
- MySQL: Relational database for all data persistence
- Jinja2 Templates: Server-side HTML rendering
- Bootstrap CSS: Responsive and modern UI components

• SYSTEM ARCHITECTURE

The Smart Library System follows a three-tier architecture consisting of the Presentation Layer (HTML, CSS, JS), the Application Layer (Python Flask), and the Data Layer (MySQL database). The Flask backend handles all HTTP requests, processes business logic, and communicates with the MySQL database via SQL queries. The frontend renders data received from the server and provides interactive forms for user inputs.

Role-based authentication is implemented using Flask sessions, ensuring that only authorized users can access specific modules. The Admin has unrestricted access, while Librarians and Members have controlled access based on their roles.

• MODULE DESCRIPTION

Authentication Module

- Login Page with role-based redirection
- Registration for new members
- Session management and logout

Admin Module

- Dashboard with system overview and statistics
- Manage librarians and members
- View all transactions and generate reports
- System configuration and settings

Librarian Module

- Add, update, delete, and search books in inventory
- Process book issue and return requests
- Manage member records
- View overdue books and calculate fines

Member Module

- Browse and search the book catalog
- View book availability status
- Track personal borrowing history
- View pending fines and due dates

Inventory Management Module

- Real-time book stock tracking
- Category and genre-based book organization
- Book availability and reservation status

• DATABASE DESIGN

The MySQL database consists of the following key tables:

- users: Stores user credentials and role information
- books: Stores book details such as title, author, ISBN, genre, and quantity
- members: Stores member profile and contact information
- transactions: Records all book issue and return transactions
- fines: Tracks overdue fines per member per transaction
- categories: Stores book categories and genres

Relationships are established using foreign keys to maintain data integrity across all tables. Indexes are applied on frequently searched columns such as book title, ISBN, and member ID to optimize query performance.

• RESULT AND OUTPUT

The Smart Library Inventory & Access Management System successfully delivers a fully functional web application with role-based access control. All modules including inventory management, member management, book transactions, and fine tracking operate as expected. The system provides real-time data updates, efficient search functionality, and clean, responsive user interfaces.

ADVANTAGES

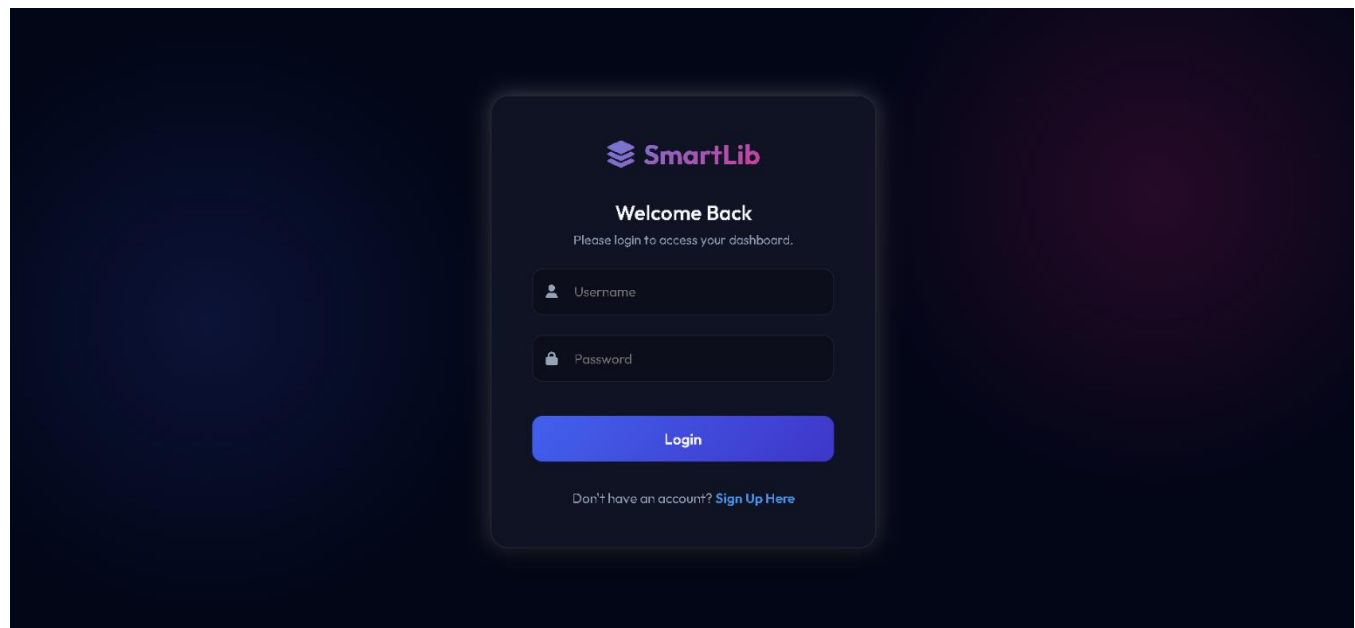
- Eliminates manual record-keeping and reduces human error
- Real-time inventory tracking and availability status
- Role-based access ensures data security and privacy
- Automated fine calculation reduces administrative overhead
- Centralized database enables easy data retrieval and reporting
- Responsive interface accessible from any device

LIMITATIONS

- No RFID or barcode scanner integration
- No mobile application support currently
- No email or SMS notification system for due dates
- Limited to single-branch library operations

FUTURE ENHANCEMENTS

- RFID/barcode scanner integration for automated book tracking
- Mobile application using React Native
- Email and SMS reminders for due dates and fines
- Multi-branch library support
- AI-based book recommendation system for members
- Integration with Google Books API for book metadata



MAIN DASHBOARD

Home

Catalog

Members

LIBRARY OPERATIONS

Add New Book

Register Member

Issue Book

Return Book

ADMINISTRATION

Admin User

Library Manager

Logout

Dashboard Overview

Smart Education Initiative - AP State Library Network

English



Search title, ISBN, or subject...



Scan

2,543
TOTAL ASSETS

1,840
ACTIVE INVENTORY

856
REGISTERED MEMBERS

42
PENDING RETURNS

Active Borrowings

View All

- The Great Gatsby**
Issued to: John Doe (Mem ID: M-101) Due in 2 days
- Clean Architecture**
Issued to: Alice Smith (Mem ID: M-023) Due in 14 days
- Introduction to Algorithms**
Issued to: Bob Johnson (Mem ID: M-499) Due in 20 days

System Alerts

Overdue Notice
3 members have books overdue by more than 7 days.

System Status
All services running perfectly. Database backup completed 2 hours ago.

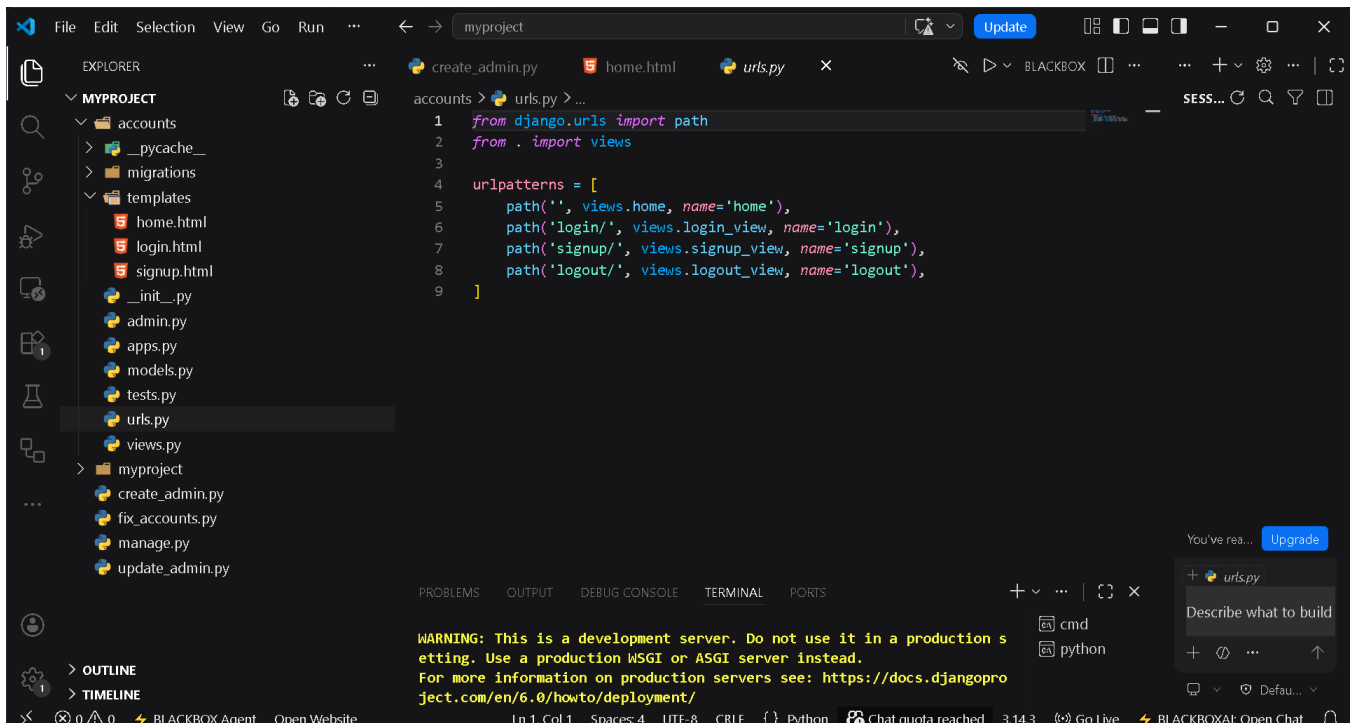
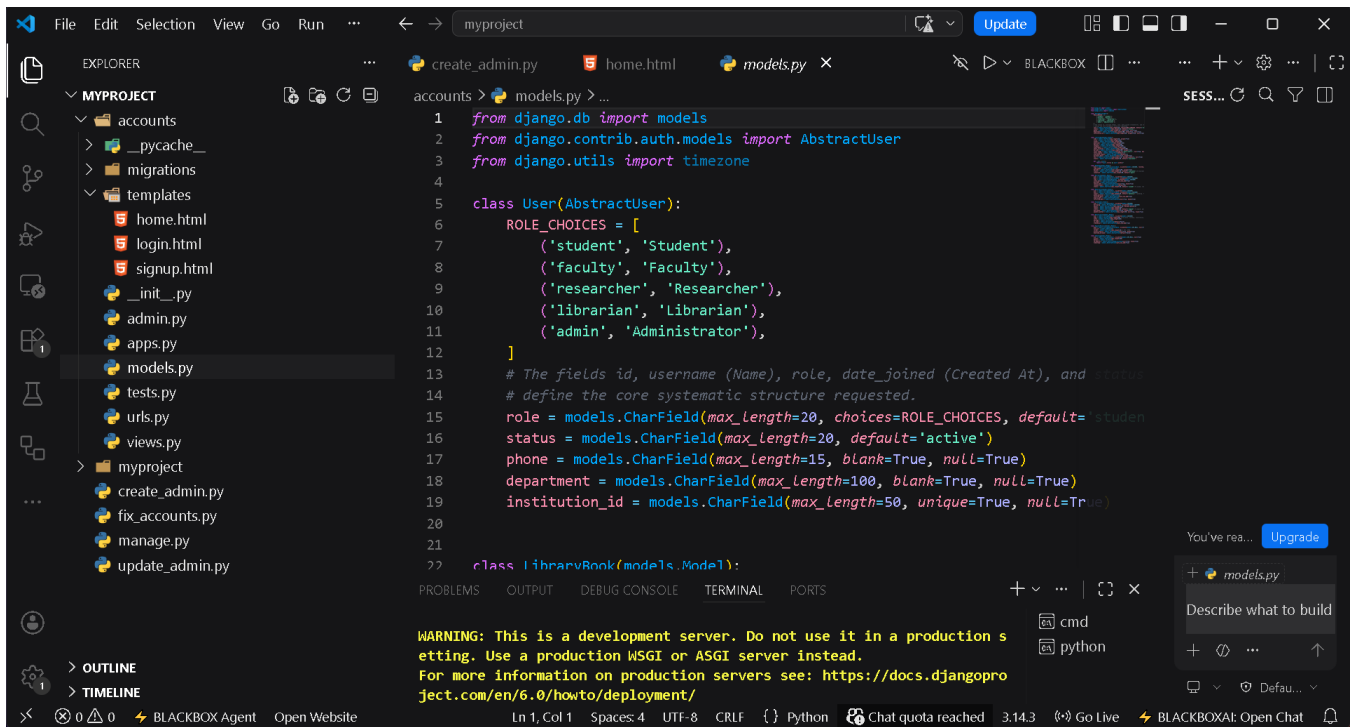
Digital Resource Hub

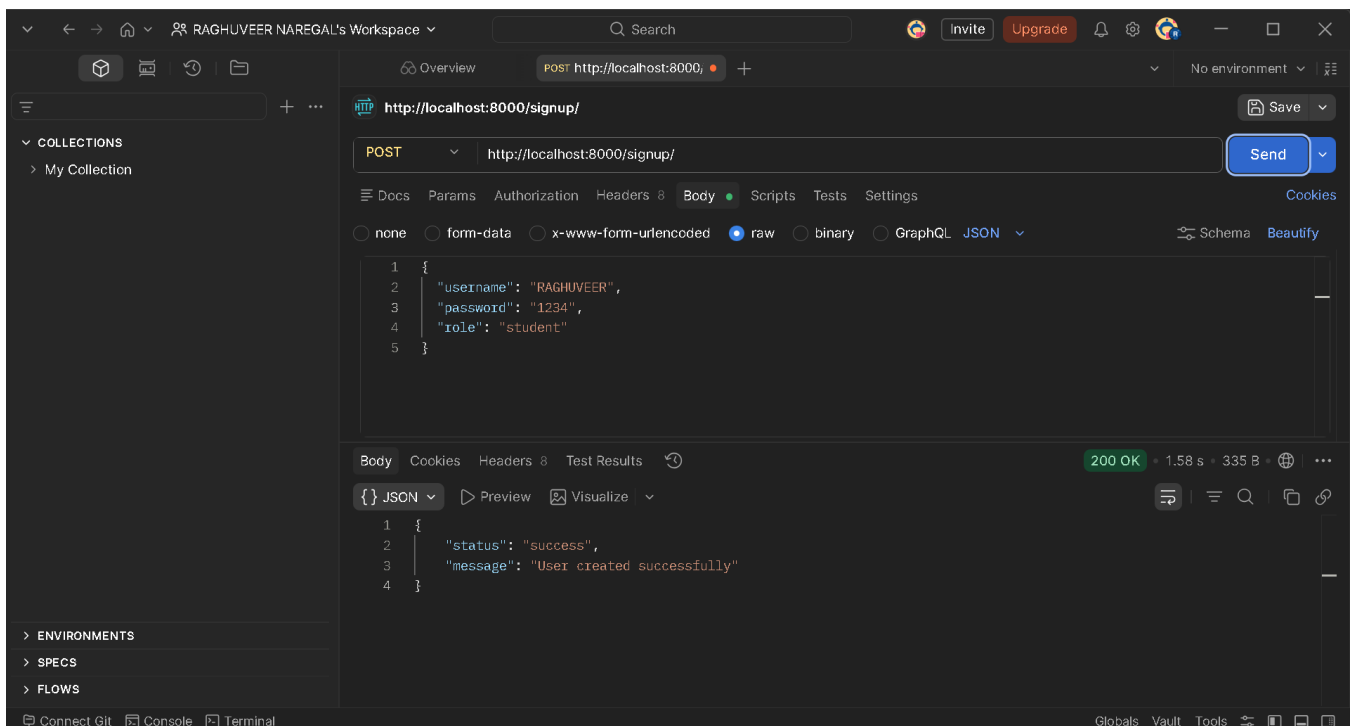
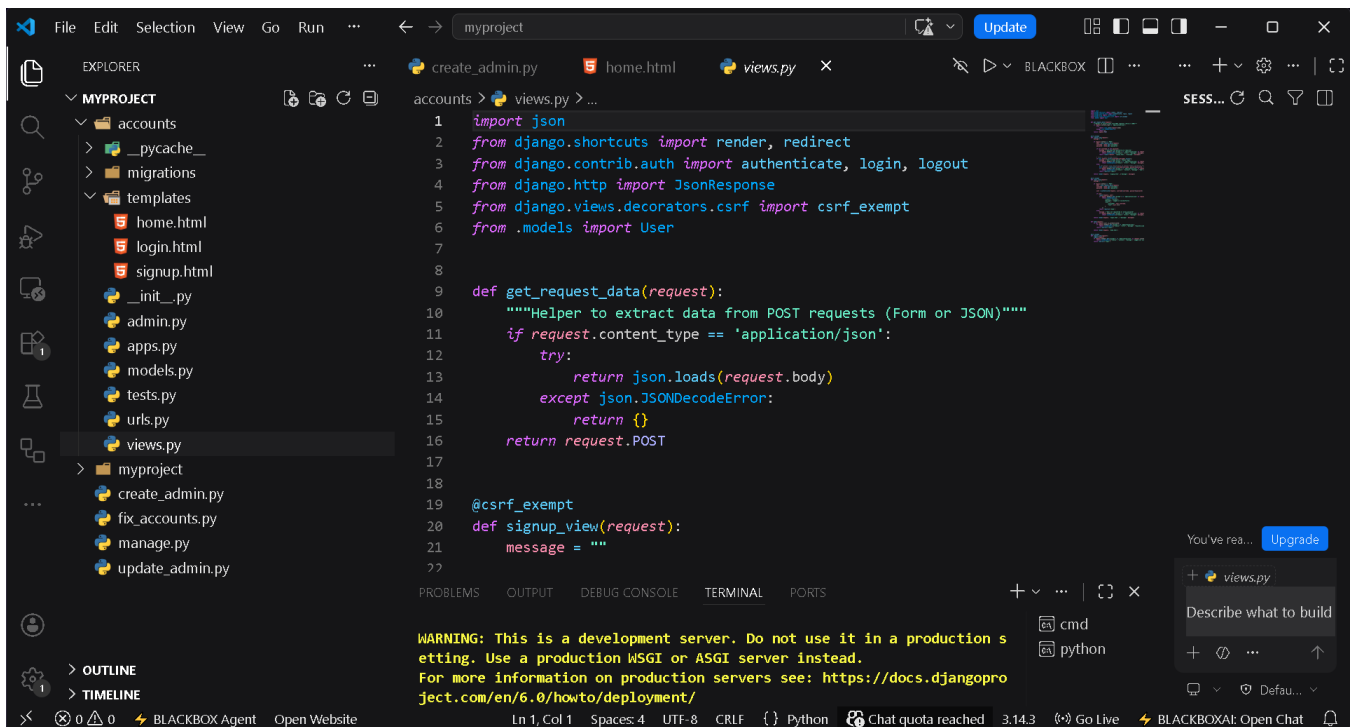
Browse E-Library

AP History Vol I
PDF - 12.4 MB

Modern Physics
EPUB - Interactive

```
File Edit Selection View Go Run ... myproject Update
EXPLORER
  MYPROJECT
    accounts
      __pycache__
      migrations
      templates
        home.html
        login.html
        signup.html
      __init__.py
      admin.py
      apps.py
      models.py
      tests.py
      urls.py
      views.py
    myproject
      create_admin.py
      fix_accounts.py
      manage.py
      update_admin.py
  create_admin.py home.html admin.py x
accounts > admin.py > ...
1 from django.contrib import admin
2 from django.contrib.auth.admin import UserAdmin
3 from django.contrib.auth.models import Permission
4 from django.contrib.contenttypes.models import ContentType
5 from django.contrib.sessions.models import Session
6 from django.contrib.admin.models import LogEntry
7 from .models import User
8
9 # Register Custom User with its special fields
10 @admin.register(User)
11 class CustomUserAdmin(UserAdmin):
12     # Overriding fieldsets to remove password and other technical fields
13     fieldsets = (
14         (None, {'fields': ('username', 'role', 'status')}),
15         ('Personal Info', {'fields': ('first_name', 'last_name', 'email', 'phone', 'is_active', 'is_staff', 'is_superuser', 'is_superuser', 'last_login', 'date_joined')}),
16         ('Permissions', {'fields': ('is_active', 'is_staff', 'is_superuser', 'is_superuser', 'last_login', 'date_joined')}),
17     )
18
19 add_fieldsets = (
20     (None, {
21         'classes': ('wide',),
22         'fields': ('username', 'role', 'status', 'password'), # Password and
23     )
24 )
25
26 WARNING: This is a development server. Do not use it in a production s
27 etting. Use a production WSGI or ASGI server instead.
28 For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/
29
30 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
31 You've rea... Upgrade
32 + admin.py
33 Describe what to build
34 + ...
35 cmd
36 python
37 Chat quota reached 3.143 Go Live BLACKBOXAI: Open Chat
```






```
Command Prompt x + ~
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 24
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use smart_library_db;
Database changed
mysql> show tables;
+-----+
| Tables_in_smart_library_db |
+-----+
| accounts_auditlog          |
| accounts_bookcopy          |
| accounts_borrowing         |
| accounts_digitalresource   |
| accounts_fine              |
| accounts_librarybook       |
| accounts_notification      |
| accounts_reservation       |
| accounts_user              |
| accounts_user_groups       |
| accounts_user_user_permissions |
| auth_group                 |
| auth_group_permissions     |
| auth_permission            |
| django_admin_log           |
| django_content_type        |
| django_migrations          |
| django_session             |
+-----+
18 rows in set (0.01 sec)

mysql> select id, username, role, last_login, status from accounts_user;
+----+-----+-----+-----+-----+
| id | username                               | role   | last_login           | status |
+----+-----+-----+-----+-----+
| 1  | raghuveeranaregal@gmail.com          | admin  | 2026-04-22 07:19:40.744896 | active |
| 2  | RAGHUVEER                           | student | 2026-04-25 06:21:49.804225 | active |
+----+-----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql>
```

• CONCLUSION

The Smart Library Inventory & Access Management System successfully demonstrates the design and implementation of a full-stack web application using Python, HTML, CSS, JavaScript, and MySQL. The project provides a practical solution to the challenges of manual library management by automating key workflows and providing role-based access to all stakeholders.

This project has provided valuable hands-on experience with full-stack web development technologies and database design principles. It serves as a solid foundation for building enterprise-level library management solutions with advanced features in the future.

• REFERENCES

- Flask Documentation - <https://flask.palletsprojects.com>
- MySQL Reference Manual - <https://dev.mysql.com/doc/>
- W3Schools HTML/CSS/JS Tutorials - <https://www.w3schools.com>
- Bootstrap Documentation - <https://getbootstrap.com/docs>
- Python Official Documentation - <https://docs.python.org/3/>